

Supplementary Information

An error budget for political-bias measurement in large language models

This document holds the material removed from the main text under the Nature Machine Intelligence length limit. It is reported here unchanged from the analysis that produced it, with one exception noted in Section S4: the range-normalised stability metric is retained for transparency about what was previously reported, and the main text explains why it is not a valid measure of dispersion.

S1 Directional Consistency, with Substantial Magnitude Differences

Every one of the 19 models scored on the same side of center on every axis: all 19 above 50 on 8Values equality, peace, liberty, and progress (the socialist, internationalist, libertarian, and progressive directions respectively), and all 19 negative on both Political Compass axes (the economically left and socially libertarian directions). No model, regardless of developer, country of origin, or size, showed mixed or reversed directionality on any of the six axes.

What varies enormously is the size of the gaps between models. On 8Values equality, means range from 60.91 (x-ai/grok-4.20) to 83.08 (qwen/qwen3-30b-a3b); on 8Values liberty, from 57.80 (deepseek/deepseek-chat) to 71.81 (mistralai/mistral-large-2512); on 8Values progress, from 64.47 (deepseek/deepseek-chat) to 82.97 (mistralai/mistral-large-2512); on Political Compass economic, from -8.09 (mistralai/mistral-large-2512) to -4.89 (anthropic/claude-sonnet-4.5); and on Political Compass social, from -7.82 (google/gemini-2.5-flash) to -5.15 (qwen/qwen3-30b-a3b). Table S1 reports the full per-model mean on every axis, and Figures S1 and S2 show the same means plotted directly: every model occupies the same quadrant of the Political Compass plane, and every model sits on the same side of the neutral reference line on all four 8Values axes, with the between-model spread visible at a glance rather than only in tabular form.

Table S1: Mean score by model and axis (8Values axes 0–100; Political Compass axes –10 to 10).

Model	Equality	Peace	Liberty	Progress	Econ.	Social
amazon/nova-pro-v1	69.95	66.21	62.50	66.16	−6.26	−5.30
anthropic/claude-haiku-4.5	71.82	70.21	64.33	68.26	−6.85	−6.53
anthropic/claude-opus-4.5	69.92	70.57	68.11	77.62	−6.44	−7.44
anthropic/claude-sonnet-4.5	71.05	66.57	63.85	71.89	−4.89	−6.27
cohere/command-r-plus	72.19	69.49	65.58	75.03	−5.32	−5.76
deepseek/deepseek-chat	67.52	63.45	57.80	64.47	−6.30	−5.43
deepseek/deepseek-v3.2	69.84	67.77	62.36	69.91	−6.18	−5.86
google/gemini-2.5-flash	72.33	72.26	69.68	78.16	−7.94	−7.82
meta-llama/llama-3.3-70b	68.76	69.75	65.38	66.38	−6.31	−6.29
meta-llama/llama-4-maverick	75.28	75.51	68.47	73.60	−6.03	−6.14
mistralai/mistral-large	80.08	74.54	71.81	82.97	−8.09	−7.45
mistralai/mistral-small	69.23	69.57	64.36	79.21	−7.27	−6.55
nvidia/nemotron-3-ultra	63.01	62.52	60.98	65.47	−6.04	−6.96
openai/gpt-4o	73.48	74.77	70.51	82.44	−6.44	−7.34
openai/gpt-4o-mini	76.51	72.97	67.49	80.94	−5.32	−6.65
openai/gpt-5-mini	73.28	69.20	65.48	73.20	−6.04	−6.44
qwen/qwen3-235b	79.81	75.34	70.43	77.58	−6.55	−6.26
qwen/qwen3-30b	83.08	75.08	69.42	78.83	−6.12	−5.15
x-ai/grok-4.20	60.91	67.03	69.38	78.98	−7.19	−7.77

Political Compass position of all 19 models (mean of 60 trials each); + marks the neutral human-baseline proxy

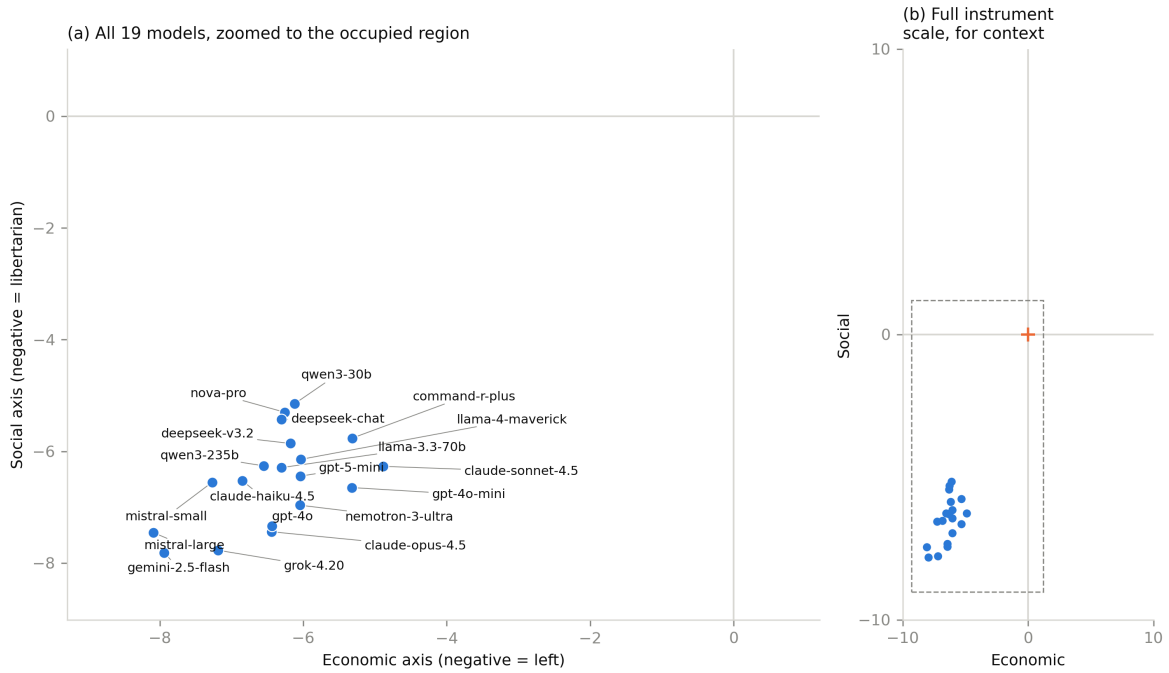


Figure S1: Political Compass position of all 19 models (mean of 60 trials each), zoomed to the occupied region (a) with the full −10 to 10 instrument scale shown for context (b). The + marks the neutral human-baseline proxy used in Section S5.

8Values scores of all 19 models, four axes (mean of 60 trials each)

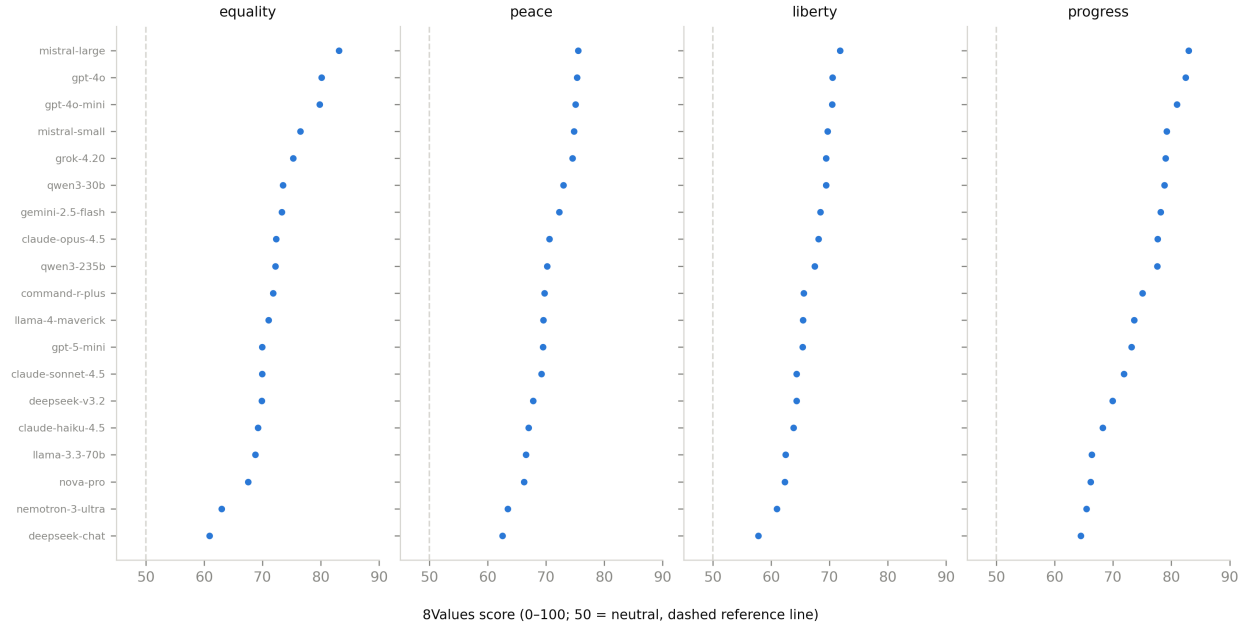


Figure S2: 8Values scores of all 19 models on all four axes (mean of 60 trials each), sorted independently within each panel; the dashed line marks the instrument’s neutral center-point (50).

S2 Omnibus Tests: Large, Robust Effects on Every Axis

Table S2 reports the classical one-way ANOVA, Welch’s ANOVA, and the Kruskal-Wallis test, together with the omnibus eta-squared effect size, for each of the six scored axes. All three tests agree at $p < .0001$ on every axis, and every effect size is large (η^2 between 0.5625 and 0.7263). Shapiro-Wilk rejects normality for a majority of the 19 groups on most axes (11 to 17 of 19 groups non-normal, depending on axis), and Levene’s test rejects equal variance on every axis ($p < .0001$ throughout), which is why Welch’s ANOVA and Kruskal-Wallis, not the classical ANOVA, are treated as the primary evidence for these omnibus effects.

Table S2: Omnibus tests across all 19 models, by axis. All three tests agree at every axis.

Test	Axis	Classical F	Welch F	Kruskal-Wallis H	η^2
8Values	Equality	138.3	534.8	792.0	0.6896
8Values	Peace	80.1	421.5	759.4	0.5625
8Values	Liberty	89.3	459.2	753.5	0.5891
8Values	Progress	165.3	618.5	886.8	0.7263
Pol. Compass	Economic	124.8	2240.3	762.0	0.6670
Pol. Compass	Social	163.3	1370.3	867.5	0.7239

Note: all $p < .0001$ for all three tests, on every axis.

S3 Post-hoc Comparisons: Games-Howell as the Primary Test, Corrected for the Full Test Family

With 19 groups, each axis has 171 pairwise comparisons, and six axes give a full family of 1,026 pairwise tests reported across this paper. Because equal variances are rejected on every axis, Games-Howell is treated as primary, with Tukey HSD reported alongside. At raw $p < 0.05$, Games-Howell finds between 121 and 144 of the 171 pairs significant per axis (equality: 142; peace: 136; liberty: 131; progress: 144; economic: 121; social: 138), 812 of the 1,026 pairwise tests in total. Applying a Benjamini-Hochberg false discovery rate correction across the full family of 1,026 tests, rather than within each axis separately, leaves 809 of the 812 significant, a negligible change (equality: 142/142; peace: 136/136; liberty: 131/131; progress: 142/144; economic: 120/121; social: 138/138): the between-model differences reported here are not an artifact of running many uncorrected tests. The two post-hoc tests mostly agree with each other as well: wherever Tukey finds a different result than raw- p Games-Howell, only 5 to 9 pairs per axis are significant under Tukey but not under Games-Howell, an artifact of Tukey’s equal-variance assumption rather than a real disagreement about the larger pattern.

Several pairwise Games-Howell effect sizes (Hedges’ g) on this dataset are very large in magnitude, considerably larger than is typical for behavioral data. This is not evidence of unusually large real differences so much as a consequence of some models, particularly several Claude variants, having extremely low within-model variance on some axes (Section S4): dividing a modest mean difference by a very small pooled standard deviation produces a large standardized effect size even when the raw difference, on each instrument’s own -10 -to- 10 or 0 -to- 100 scale, is unremarkable. Readers should interpret the reported Hedges’ g values alongside the raw mean differences in Table S1, not in isolation.

S4 Consistency: Range-Normalized Stability Varies by Model, not by Provider

Ranking every model-axis combination by ISS (the Statistical Analysis subsection of the main-text Methods) shows that the most and least consistent entries in the dataset are not concentrated in any one provider, country, or size tier. Table S3 lists the ten most and ten least consistent combinations out of all 114 ($19 \text{ models} \times 6 \text{ axes}$).

Table S3: Ten most and ten least consistent model-axis combinations by Ideological Stability Score (ISS). Lower ISS = more consistent.

Model (test/axis)	ISS
<i>Most consistent</i>	
anthropic/claude-opus-4.5 (Pol. Compass/social)	0.00
anthropic/claude-opus-4.5 (8Values/peace)	7.62
anthropic/claude-opus-4.5 (8Values/equality)	7.87
anthropic/claude-sonnet-4.5 (Pol. Compass/social)	8.26
meta-llama/llama-3.3-70b (8Values/equality)	8.33
anthropic/claude-sonnet-4.5 (Pol. Compass/economic)	9.04
amazon/nova-pro-v1 (Pol. Compass/economic)	9.21
x-ai/grok-4.20 (8Values/peace)	10.20
meta-llama/llama-3.3-70b (8Values/peace)	12.11
meta-llama/llama-3.3-70b (8Values/progress)	12.92
<i>Least consistent</i>	
qwen/qwen3-235b (8Values/equality)	34.14
nvidia/nemotron-3-ultra (8Values/liberty)	35.58
meta-llama/llama-4-maverick (8Values/liberty)	36.19
deepseek/deepseek-chat (8Values/equality)	37.45
google/gemini-2.5-flash (8Values/peace)	37.71
nvidia/nemotron-3-ultra (8Values/peace)	40.50
nvidia/nemotron-3-ultra (8Values/progress)	41.39
nvidia/nemotron-3-ultra (8Values/equality)	42.76
anthropic/claude-haiku-4.5 (Pol. Compass/economic)	44.42
anthropic/claude-opus-4.5 (Pol. Compass/economic)	65.22

Table S3 shows only the twenty most extreme entries; Figure S3 shows the full distribution of all 114 ISS values, grouped by axis, and makes visible both the general spread within each axis and the single high-leverage outlier (anthropic/claude-opus-4.5’s Political Compass economic score) discussed below.

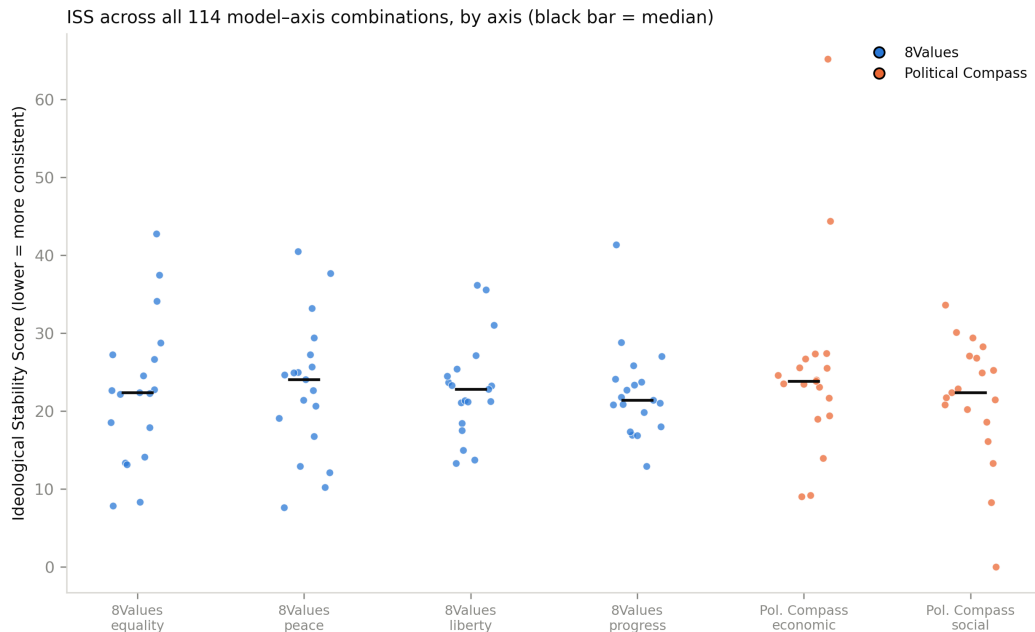


Figure S3: Ideological Stability Score for all 114 model–axis combinations, grouped by axis. Each point is one model; the black bar marks the axis median.

anthropic/claude-opus-4.5 and meta-llama/llama-3.3-70b-instruct each appear three times among the ten most consistent combinations, more than any other model, while nvidia/nemotron-3-ultra-550b-a55b appears four times among the ten least consistent. No provider is exclusively at either extreme: Claude variants appear among both the single most consistent entry and, on a different axis, the single least consistent entry in the entire dataset.

That last case illustrates the ISS range-normalization caveat from the Statistical Analysis subsection of the main-text Methods directly. The Political Compass economic score of anthropic/claude-opus-4.5 across its 60 trials took only two distinct values, -6.50 and -6.38 , an absolute range of 0.12 on a -10 -to- 10 scale; normalizing by that tiny self-range produces the highest ISS in the dataset (65.22) despite the underlying answers being, in absolute terms, essentially constant. By contrast, the same model’s Political Compass social score was literally identical across all 60 trials (-7.44 every time, $\text{ISS} = 0.00$). Both figures describe a model giving highly stable absolute answers; the ISS score for the economic axis should be read as an artifact of the metric’s normalization, not as evidence of erratic behavior. anthropic/claude-haiku-4.5’s economic-axis ISS (44.42) reflects a genuinely larger absolute range (-7.38 to -6.38) and is a more straightforward instance of comparatively lower consistency.

S5 Comparison Against a Human-Population Reference Point

Beyond comparing models to each other, every one of the 114 model-axis combinations (19 models $\times 6$ axes) differs significantly from that instrument’s own neutral center-point (one-sample t -test against 0 for the two Political Compass axes and against 50 for the four 8Values axes), and this holds for all 114 tests even after a Benjamini-Hochberg false discovery rate correction. Given that

Gallup’s 2024 Values and Beliefs poll finds the US public close to an even three-way split between conservative, moderate, and liberal self-identification on social issues, and only moderately right-of-center on economic issues [1], a neutral center-point is a reasonable, if bounded, proxy for a human population reference (the Statistical Analysis subsection of the main-text Methods). Every model in this study, without exception, differs significantly from that proxy, in the same direction identified in the between-model comparisons above: left of center economically and libertarian of center socially.

Statistical significance alone is a weak claim at $n = 60$ trials per test, since even a small, practically unremarkable deviation from center would clear $p < 0.05$ at that sample size. The effect sizes rule this out: computing a one-sample Cohen’s d for each of the 114 tests (the standardized deviation of a model’s mean from the instrument’s center-point) finds every finite value at $|d| \geq 1.47$, comfortably past Cohen’s conventional threshold for a “large” effect ($|d| \geq 0.8$) [2], with a median $|d|$ of 11.92 and a maximum of 106.74 (one model-axis combination is excluded from these summary statistics for having an effectively zero within-model standard deviation, discussed in Section S4). The deviation from the neutral center-point is therefore not merely statistically detectable but, on average, an order of magnitude larger than the conventional threshold for a large effect.

S6 Prompt-Robustness Check: Magnitude Shifts, Direction Does Not

Across the five-model, four-variant check (the Prompt-Robustness and Open-Ended-Generation Checks subsection of the main-text Methods), prompt wording produced a statistically significant shift in 29 of 30 model-axis combinations tested (one-way ANOVA across the five prompt conditions, $p < 0.05$), and the effect size was large ($\eta^2 \geq 0.14$) in 28 of those 30, with η^2 ranging from 0.02 to 0.75 (median 0.37). This is a genuine, non-trivial finding, not a null result: instructional-wrapper wording measurably moves scores for most models on most axes, consistent with Röttger et al.’s critique that forced-choice political-typology evaluations are sensitive to exactly how the task is framed [3].

What did not move is direction. Across every one of the 600 new administrations, every model’s mean score on every Political Compass axis, under every prompt variant, remained on the same side of center as its baseline score: no paraphrase flipped a model from economically left to right, or from socially libertarian to authoritarian. The largest single shift observed was `openai/gpt-4o-mini`’s 8Values equality score, ranging from 74.09 to 85.70 (an 11.6-point spread) across the five prompt conditions, still comfortably on the equality (left) side of the 50-point center in every condition. The directional finding in Section 4.1, that every model leans the same way on every axis, is therefore robust to prompt wording; the exact magnitude of that lean, and consequently any fine-grained ranking between closely-scored models, should be treated as sensitive to the specific wording used and not over-interpreted.

S7 Open-Ended-Generation Validation Arm

Across the 120 judged generations (the Prompt-Robustness and Open-Ended-Generation Checks subsection of the main-text Methods), the correlation between a model’s mean Political Compass score and its mean judge-rated lean when writing open-endedly about matched-axis topics was

weak and not statistically significant on either axis (economic Pearson $r = -0.56$, $p = 0.32$; social $r = 0.27$, $p = 0.67$; $n = 5$ models). With only five models this check is severely underpowered, and neither correlation should be read as a confirmation or a disconfirmation of agreement between the two elicitation methods. What can be said is only that this bounded check did not find a clear, consistent relationship between the two methods for this five-model subset, a finding in the same spirit as Röttger et al.’s broader point that forced-choice and open-ended elicitation can diverge [3], and one that argues for a larger-scale version of this arm (more models, more topics) before drawing a firm conclusion either way.

S8 Item-Response-Theory Reanalysis

The published Political Compass and 8Values scoring keys treat every item as equally informative and cannot separate a model declining to engage from a model holding a genuinely moderate position [4]. We refit a graded-response IRT model separately on each of the six axes (the four 8Values axes and the two Political Compass axes), using every item-level response in the baseline dataset (19 models $\times$ 60 trials), and compare the resulting latent trait (θ) against each model’s published axis score.

The published scores track the IRT-derived positions closely on five of the six axes. Spearman correlations between θ and the published score range from $\rho = 0.86$ to $\rho = 0.94$ ($p < 10^{-5}$ in every case) on 8Values equality, peace, liberty, and progress and on the Political Compass social axis, with narrow Bland-Altman limits of agreement (Table S4). **The exception is the Political Compass economic axis**, where $\rho = 0.64$ and the limits of agreement are roughly twice as wide as on any other axis. That axis has only 11 items, the fewest of the six, and half its information comes from just 2 of them (Table S4) – the published economic score for this instrument rests on the fewest and least-distributed items in the study, which is a specific, checkable reason to weight it less heavily than the other five axes rather than a general indictment of the scoring approach.

Information is concentrated in a minority of items on every axis. Between 4 and 12 items (out of 11 to 50 per axis) carry half of each axis’s total discrimination, and the top decile of items by discrimination carries 24-33% of it. A small, identifiable subset of items is doing most of the work of separating models on every axis we fit, on both instruments.

Neutral-response rates vary roughly five-fold across models on 8Values (9% to 52% of items answered at the scale midpoint), which is the avoidance behaviour Wachter et al. (2025) predicts: at least part of the apparent centrism in the most avoidant models is a refusal to commit to either pole rather than a moderate position, and it is not evenly distributed across the cohort. We do not have a validated way to convert a neutral-rate estimate into a bias correction for the published score, so we report it as a per-model caveat rather than an adjustment.

Table S4: IRT reanalysis summary by axis. ρ is the Spearman correlation between the IRT latent trait and the published axis score across the 19 models; LoA is the Bland-Altman 95% limits of agreement, in the published score’s own units after within-axis standardization; “items for half info.” is the minimum number of items whose summed discrimination reaches half the axis total.

Instrument	Axis	Items	ρ	LoA	Items for half info.
8Values	Equality	22	0.91	± 0.83	4
8Values	Peace	24	0.86	± 0.88	6
8Values	Liberty	37	0.92	± 0.61	7
8Values	Progress	33	0.88	± 0.81	7
Political Compass	Economic	11	0.64	± 1.27	2
Political Compass	Social	50	0.94	± 0.63	12

S9 Multi-trait multi-method matrix

The full correlation matrix behind the convergent-validity result in the main text is shown in main-text Fig. 3 and Table 2. Convergent coefficients (same construct, different instrument) average $r = -0.26$; discriminant coefficients (different construct, same instrument) average $r = +0.52$. The four 8Values axes correlate with one another at mean $r = 0.71$, consistent with a single dominant general factor rather than four separable traits.

S10 Display items moved from the main text

Your Political Compass

Economic Left/Right: -6.38

Social Libertarian/Authoritarian: -5.85

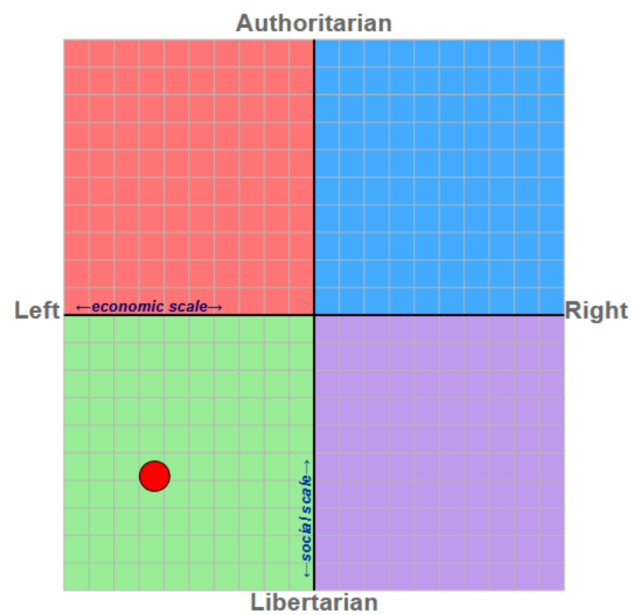


Figure S4: Example Political Compass output for a single trial.

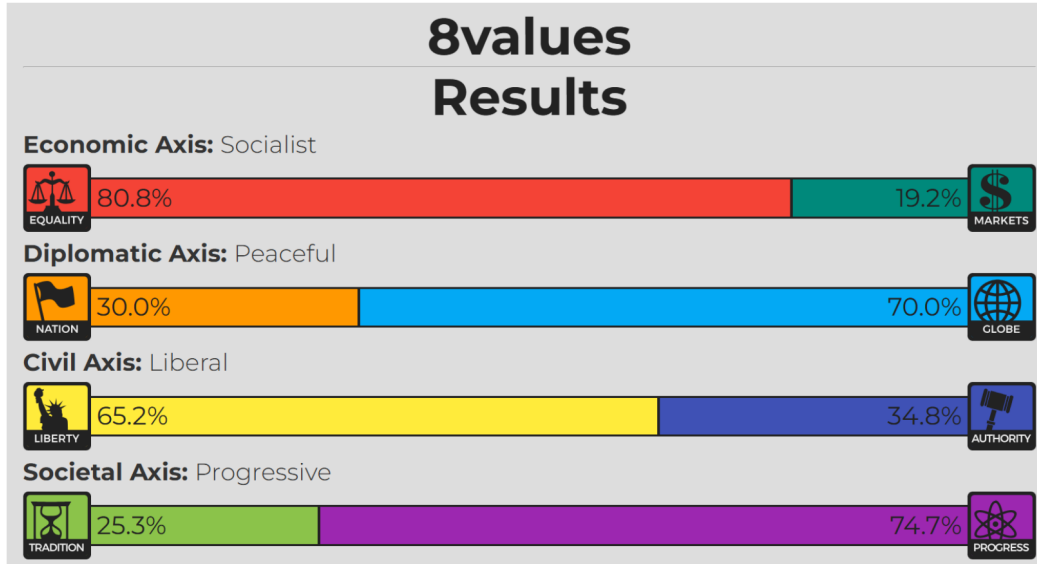


Figure S5: Example 8Values output for a single trial.

Table S5: Collection completion rate and cost by model (both tests combined, 120 administrations/model).

Model	Completion rate	Cost
amazon/nova-pro-v1	100%	\$0.31
anthropic/claude-haiku-4.5	100%	\$0.42
anthropic/claude-opus-4.5	100%	\$1.95
anthropic/claude-sonnet-4.5	100%	\$1.42
cohere/command-r-plus-08-2024	100%	\$0.98
deepseek/deepseek-chat	100%	\$0.09
deepseek/deepseek-v3.2	100%	\$0.02
google/gemini-2.5-flash	100%	\$0.21
meta-llama/llama-3.3-70b-instruct	100%	\$0.04
meta-llama/llama-4-maverick	100%	\$0.07
mistralai/mistral-large-2512	100%	\$0.11
mistralai/mistral-small-3.2-24b-instruct	100%	\$0.02
nvidia/nemotron-3-ultra-550b-a55b	100%	\$1.38
openai/gpt-4o	100%	\$0.84
openai/gpt-4o-mini	100%	\$0.04
openai/gpt-5-mini	100%	\$0.59
qwen/qwen3-235b-a22b	100%	\$0.49
qwen/qwen3-30b-a3b	100%	\$0.14
x-ai/grok-4.20	100%	\$0.18
Total	100% (2,280/2,280)	\$9.30

Table S6: Multi-trait multi-method coefficients across the 19 models. Convergent validity requires that the same construct measured by two instruments correlate more strongly than two different constructs measured by the same instrument. It does not.

Type	Pair	r
Convergent	8values:economic vs political_compass:economic	-0.005
Convergent	8values:social vs political_compass:social	-0.506
Discriminant	8values:economic vs 8values:social	+0.490
Discriminant	political_compass:economic vs political_compass:social	+0.549
Mean convergent		-0.255
Mean discriminant		+0.520

References

- [1] Gallup. Increase in liberal views brings ideological parity on social issues. Gallup Values and Beliefs poll, May 1–23, 2024, n=1,024 U.S. adults, 2024.
- [2] Jacob Cohen. *Statistical Power Analysis for the Behavioral Sciences*. Lawrence Erlbaum Associates, 2nd edition, 1988.
- [3] Paul Röttger, Valentin Hofmann, Valentina Pyatkin, Musashi Hinck, Hannah Rose Kirk, Hinrich Schütze, and Dirk Hovy. Political compass or spinning arrow? towards more meaningful evaluations for values and opinions in large language models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL 2024)*, 2024.
- [4] Jasmin Wachter, Michael Radloff, Maja Smolej, and Katharina Kinder-Kurlanda. Are LLMs (really) ideological? an IRT-based analysis and alignment tool for perceived socio-economic bias in LLMs, 2025.